Dr.-Ing. Danish IQBAL

Computational Engineer

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Essen, Germany 10/03/1991

https://m-d-iqbal.github.io https://linkedin.com/in/drdanishiqbal https://scholar.google.de/citations?user=cxAW-3MAAAAJ

PROFILE

I am a Ph.D. Computational Engineer with a research background in advanced numerical methods, multi-physics simulation, and fracture mechanics. My doctoral work focused on developing and implementing the Scaled Boundary Finite Element Method (SBFEM) for coupled thermoelastic problems in engineered materials. Beyond research, I have built practical skills in software development, workflow automation, and system integration, applying them in a production environment. I bring a rare combination of research depth and hands-on engineering practice, bridging advanced simulation and numerical methods with modern software development and automation.

PROFESSIONAL EXPERIENCE

07/2025 - present

Systems Engineer

CleverBuy GmbH, Krefeld, Germany

- Designed and built an automation system that replaces time-consuming manual workflows with automated, real-time processes running across multiple connected business systems.
- Built an asynchronous job engine that continuously evaluates and applies business rules over large volumes of operational data, removing the need for manual monitoring or intervention.
- Developed API integrations connecting several external platforms, enabling reliable and automated data exchange across systems.
- Designing an AI integration layer that gives a language model structured access to the business database, enabling users to query, plan, and action changes through natural conversation.

08/2024 - 06/2025

Professional Development

- Pursued self-directed learning in full-stack development, data science, and machine learning, strengthening and expanding core engineering competencies.

08/2017 - 03/2024

Research Assistant

University of Duisburg-Essen, Essen, Germany

- Designed, developed, and implemented numerical method (SBFEM) in MATLAB for multi-physics simulation.
- Utilized ANSYS, Abaqus, and FEAP for advanced simulations, including scripting for automation and customization.
- Taught graduate courses in Structural Dynamics and programming with ANSYS, Fortran, C++, and Python.
- Supervised three master's one bachelor's theses on diverse topics in computational mechanics.
- Managed an e-learning project to digitize course content and develop tools like inverted classroom and virtual labs.

PROFESSIONAL EXPERIENCE

01/2020 - 03/2020

Visiting Researcher

Federation University, Ballarat, Australia

- Collaborated with the University of Duisburg-Essen and Federation University under a prestigious Australia-Germany research grant, advancing SBFEM for modeling thermoelastic fracture in specialized engineering materials.

01/2017 - 03/2017

Student Assistant

University of Duisburg-Essen, Essen, Germany

- Implemented a polygon-based SBFEM in FORTRAN and integrated it into the open-source software FEAPpv.

06/2015 - 10/2015

Student Worker

Mercedes-Benz Group AG, Düsseldorf, Germany

- Executed spot welding with jigs and performed automotive assembly line tasks.

10/2014 - 12/2014

Student Worker

Mercedes-Benz Group AG, Rastatt, Germany

- Carried out production tasks on automotive assembly line.

12/2013 - 12/2013

Logistics Assistant

Amazon Deutschland Services GmbH, Rheinberg, Germany

- Scanned items and prepared them for packaging.

06/2011 - 07/2011

Internship

Karachi Tools, Dies & Moulds Center, Karachi, Pakistan

- Practical training in CAD/CAM and nitriding heat treatment.

CORE COMPETENCIES

- Multi-Physics Simulation
- Numerical Methods
- Engineering Materials
- Software Development
- Workflow Automation
- Project Management

SKILLS

Numerical Methods:	• SBFEM	• FEM	• XFEM	• CFD
Simulation Tools:	• ANSYS	• Abaqus	• Gmsh	• ParaView
Programming:	• MATLAB	• Python	• Fortran	• C++
Dev. Stack:	• Django	• PostgreSQL	• REST APIs	• Git

EDUCATION

08/2017 - 07/2024

Ph.D. Computational Mechanics

University of Duisburg-Essen, Germany

- Thesis: Development of scaled boundary polygon elements for coupled thermoelastic fracture modeling.
- Grade: Magna cum laude (1.0)

EDUCATION

10/2013 - 07/2017

M.Sc. Computational Mechanics

University of Duisburg-Essen, Germany

- Thesis: Development of polygon elements based on the scaled boundary finite element method.
- Grade: Good (2.0)

01/2009 - 12/2012

B.E. Automotive Engineering

NED University of Engineering & Technology, Pakistan

- Thesis: Design and analysis of motorcycle oil pump.
- Project: Design, development, and manufacturing of Formula Student car.
- Grade: First Division (2.5)

PUBLICATIONS

Peer-Reviewed Journal Papers

M. D. Iqbal, et al. "Transient thermoelastic fracture analysis of functionally graded materials using the scaled boundary finite element method." Theoretical and Applied Fracture Mechanics, vol. 127, Oct. 2023, p. 104056.

M. D. Iqbal, et al. "Thermoelastic fracture analysis of functionally graded materials using the scaled boundary finite element method." Engineering Fracture Mechanics, vol. 264, Apr. 2022, p. 108305.

M. D. Iqbal, et al. "Development of the scaled boundary finite element method for crack propagation modeling of elastic solids subjected to coupled thermo-mechanical loads." Computer Methods in Applied Mechanics and Engineering, vol. 387, Dec. 2021, p. 114106.

E.T. Ooi, *M. D. Iqbal*, et al. "A polygon scaled boundary finite element formulation for transient coupled thermoelastic fracture problems." Engineering Fracture Mechanics, vol. 240, Dec. 2020, p. 107300.

Conference Papers

C. Birk, R. Assaf, *M. D. Iqbal*, H. Gravenkamp, "Numerische strukturmodellierung mit polygon-elementen." Baustatik-Baupraxis 15 (2024).

M. D. Iqbal, et al., "Scaled boundary finite element modelling of thermally induced crack propagation." Proceedings in Applied Mathematics and Mechanics 19 (2019).

M. D. Iqbal, C. Birk, "Numerical modelling of thermally induced crack propagation in functionally graded materials using the scaled boundary finite element method." Forschungskolloquium 2018 Grasellenbach (2018).

Theses

M. D. Iqbal, "Development of scaled boundary polygon elements for coupled thermoelastic fracture modeling." PhD Thesis, University of Duisburg-Essen (2024).

M. D. Iqbal, "Development of polygon elements based on the scaled boundary finite element method." Master Thesis, University of Duisburg-Essen (2017).

LANGUAGES

Urdu (native) | English (business fluent) | German (fluent)

Essen, 7/4/2026